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Development of a CT radiomics and clinical feature combined model for predicting early recurrence of surgical resected hepatocellular carcinoma

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Abstract

Background: Recurrence rate remains unsatisfactory among surgical resected hepatocellular carcinoma (HCC)

patients with radical resection intention, and effective surveillance methods are lacking for post-operative recurrence. **Methods:** 436 HCC patients were selected for ultimate analyses. Significant features were extracted on favorable regions of interest (ROIs) of contrast-enhanced CT (CECT) and selected by least absolute shrinkage and selection operator (LASSO) method to construct radiomics signature. A novel CT radiomics-clinicopathological prediction model was constructed to evaluate 2-year recurrence-free survival (RFS) of resected HCC. Model discrimination was evaluated by area under the receiver operating characteristic (ROC) curve, the calibrated curves and decision curve analysis. Gene expressions of CENPA, FAM83D etc. were assessed by quantitative real-time PCR (q-PCR) for 48 randomly selected HCC patients, and correlation between radiomics signature and genomic characteristics was investigated. **Results:** Radiomics signature was established based on the 20 early recurrence-related CT features. Microvascular invasion (MVI), alpha-fetoprotein (AFP), gamma-glutamyl transpeptidase to lymphocyte ratio (GLR) and radiomics signature were selected as independent predictors of early recurrence of HCC, and thereafter used to construct the novel nomogram which predicted 2-year RFS. The radiomics-clinicopathological combined model revealed favorable prediction ability with area under the ROC curve (AUC) of 0.744 and 0.821 to predict 2-year RFS in training and validation cohort. HCC patients were divided

into three risk groups, with RFS difference between groups statistically significant. Similar results were observed in AFP-negative HCC cohort. Pearson correlation analyses revealed favorable relationship between radiomics signature and BCAT1 and CENPA.

Conclusion: This study constructed a noninvasive and simple prediction model based on CT radiomics and clinical features. The novel radiomics-clinicopathological predictive model could evaluate post-operative early recurrence for HCC, and its significance remained in AFP-negative HCC.

Keywords: hepatocellular carcinoma, early recurrence, CT radiomics, prediction model, risk stratification

Introduction

Hepatocellular Carcinoma (HCC) is the most common histopathologic type of liver cancer, being the sixth most-prevalent primary cancer and third cancer-caused death worldwide¹. At present, the Barcelona Clinic Liver Cancer (BCLC) staging system is mostly applied and widely acknowledged in clinical practice, and plays a significant role in guiding treatments decision^{2,3}. For the majority of HCC patients with BCLC 0 and A stages, curative treatments including hepatic resection, liver transplantation and regional ablation are recommended. However, 5-year recurrence rate after curative treatments is still up to 70% for HCC⁴, and post-operative recurrence is one of

the important risk factors that exert great influence on tumor prognosis and long-term survival. Therefore, effective stratification before operation and early identification of HCC patients at high risk of relapse enable for more extensive surgery margin, tighter and individualized follow-up schedules and even accepting adjuvant therapy after resection, and either may benefit patients from longer survival. Unfortunately, there are lack of effective methods to monitor early recurrence of HCC after surgery, being one of the important research tasks at present.

In practice, patients who shared similar tumor stage, tumor number and size may have different clinical outcomes after operation, and drug resistance occurred during immunotherapy, which may be resulted from the various and complicated tumor microenvironment (TME) and tumor heterogeneity⁵. Some researchers attempted to learn from the underlying and subtle differences of TME from the medical images, which may figure out significant predictors of patients' prognosis and assist in making effective treatment decisions⁶⁻⁸. Based on artificial intelligence (AI) technology, which imitates the cognitive functioning of human brains, such as learning and decisive capabilities, radiomics springs up as a cross-discipline and can automatically extract a large amount of subvisual and quantitative features from the tumor and peritumor areas in medical images and stockpile them as large scale high dimensional database⁹. Machine learning (ML) and more

advanced deep learning (DL) emerging as new players in the field of medicine, are important approaches and methods to realize radiomics, and may figure out correlations between image features and tumor progression as well as patients' prognosis¹⁰⁻¹². Moreover, radio-genomics is another emerging field which investigates the relationship between image features and gene expressions, intending to offer explanations for medical images' clinical role from the perspective of gene and even tumor biological meanings¹³. Although some previous studies have established radiomics scoring models to predict risk of postoperative recurrence of HCC based on multidimensionally extracted data^{12,14,15}, the parameters of radiomics features and clinical features included in different models differed greatly, and parameters possessing more critical significance were not well-defined. Therefore, mining for important parameters and optimizing prediction models are still necessary.

In this study, we probed into the underlying and quantitative features related to shape, size, strength and texture of tumor and microenvironment from the contrast enhanced Computer Tomography (CECT) images, and constructed the novel radiomics-clinic combined prognostic model to evaluate the important image features' capability in predicting HCC patients' early recurrence risk. For HCC patients with high risk of recurrence, post-operative adjuvant therapy and more intensive monitoring examinations may be offered. At present, some biomarkers are suggested to be related

with malignant biological behavior and characteristics of malignant tumors, including HCC. Therefore, we selected 11 biomarkers which were reported by many studies¹⁶⁻¹⁸, including CENPA and FAM83D, and attempted to learn relationship between CT features and gene expression pattern, which may offer more mechanistic explanations for medical images' underlying role and significance in understanding tumor biology. This research attempted to provide noninvasive, cheap and reproducible biomarkers for HCC management, and may offer valuable information and important support for clinical decision making.

Materials and Methods

Patients Enrollment

The study was approved by the research ethics committee of Nanfang Hospital of Southern Medical University (NFEC-202305-K32), and conformed to the principles of Declaration of Helsinki. Written informed consent was waived by the ethics committee for the retrospective nature of this study. A total of 664 HCC patients underwent hepatic resection were received, and among them, 436 patients with pathologically confirmed HCC were selected for final analyses according to the inclusion and exclusion criteria shown in **Figure1**.

CECT scan was finished within one week before operation. Curative resection was defined as complete removal of tumor and no residual tumor checked by more than two examinations with no less than 4-week-interval, with pathological confirmation of negative margins. Abdominal

ultrasonography during surgery and postoperative CT scan were routinely performed to ensure complete removal of tumor. Tumor specimens were diagnosed by two independent pathologists ignorant of clinical information, and a third pathologist may be needed when the results were discordant.

Follow up surveillance

All post-operative patients received routine follow-ups every 3 months in the first year after operation and 3 to 6 months thereafter. Routine follow-up examinations include blood routine examination, hepatic function test, serum AFP level and abdominal ultrasonography. When any residual lesion or relapse were suspected, CT or MRI scan would be performed immediately. The recurrence-free survival (RFS) time was calculated from date of surgery to date of recurrence.

CECT radiomics analysis

The CECT radiomics workflow is summarized as the follows.

(1) CT Images processing and feature extraction

Each patient's CECT images were acquired in accordance with the recommendation of AASLD guideline. The standardized four-phase CECT images included the non-contrast phase, the arterial phase (AP), the portal venous phase (PVP) and the delay phase, which were obtained at timepoint of 30s, 60s and 150s after intravenous administration of the contrast medium. We chose the AP and PVP as targeted images, and intensities of all images were standardized. The CT scanners and

parameters were provided in **Table S1**.

Segmentation of the CECT images was conducted using the open-source software, 3D Slicer (version4.11, <https://www.slicer.org>). Semi-automatic segmentation method was used to outline the favorable regions of interest (ROIs) from each contiguous axial slice covering the entire tumor in both AP and PVP in the CECT images, and the segmentations were used for 3D volume of interest reconstruction. The ROI was segmented by the professional radiologist with 10 years of experience in liver imaging and subsequently reviewed by a second senior radiologist. And the divergences were finally confirmed by another specialized radiologist with more than 20 years of liver imaging experience. For patients with multiple tumors, the largest one was selected and fully contoured in 3D.

These delineated images were reconstructed and voxel size was resampled to $1*1*1\text{ mm}^3$ to adjust different voxel sizes. Radiomics features were extracted separately for each patient using the open-source Pyradiomics package (version2.12; <http://pyradiomics.readthedocs.io/en/2.1.2/>). And CT image processing was conducted using the Python software (version3.7, <https://www.python.org>) and the open-source software Simple-ITK package.

(2) Construction of radiomics signature (rad-score)

Feature value was standardized using z-score derived from the training cohort ($z = (x - \mu) / \sigma$). The early recurrence-related radiomics features were selected by least absolute shrinkage and selection operator (LASSO)

method with 10-fold-cross-validation in the training cohort and then applied to the validation cohort. These optimal features multiplied by their corresponding coefficients were then applied to construct the radiomics signature via linear combination, and each patient's point of radiomics signature was defined as radiomics score (rad-score).

Radiomics-clinic prediction model development and validation

Univariate Cox regression analysis was performed to assess the performance of a single variable for evaluating post-operative recurrence. Variables with p -value < 0.05 were then incorporated into the multivariate Cox regression analysis. Significant variables were selected by backward stepwise elimination with the Akaike Information Criteria (AIC) as the stopping rule in the multivariate Cox regression, and a novel CT radiomics and clinical feature combined prediction model was constructed to evaluate 2 year-RFS of resected HCC patients, and depicted in form of nomogram.

Model discrimination was evaluated by area under the receiver operating characteristic (ROC) curve (AUC) and the calibration curves in both training and validation cohorts, and decision curve analysis (DCA) was applied to determine potential risk threshold for clinical decision. Finally, HCC patients were differently grouped into three risk groups according to their risk scores calculated by nomogram.

Correlation analysis between CT features and

genomic characteristics

Among all enrolled HCC patients, 48 individuals were selected randomly and the tumor and peritumoral paired tissues were collected after surgery. Quantitative real-time PCR (q-PCR) analysis was used to evaluate relative expression of gene transcriptional level, by applying the Hieff UNICON® Universal Blue qPCR SYBR Green Master Mix (YEASEN) on Roche LightCycler® 480 II machine. Fold changes of relative gene expression were calculated by $2^{-\Delta\Delta C_t}$. A total of 11 biomarkers including CENPA, FAM83D, RECQL4, KPNA2, SGOL1, BCAT1, POLD1, SPC24, KIF18A, CKAP5 and DLGAP5 were analyzed, and their corresponding primer sequences and more detailed q-PCR protocol were detailed in Supplementary **Table S2**. Pearson correlation analyses were conducted to determine the association between radiomics signature and gene expression level. Absolute value of Pearson Correlation Coefficient (PCC) greater than 0.3 were considered statistically significant ($p < 0.01$).

Statistical analysis

The statistical analyses were performed using software R (version 4.11, <https://www.r-project.org/>). Continuous variables were expressed as mean $\pm$ standard deviation (SD) or median with interquartile range (IQR), respectively. Categorical variables were compared using Pearson χ^2 test or Fisher exact probability test, while continuous variables were compared using the Student's t-test or Mann-Whitney U test. RFS was estimated by

Kaplan-Meier method and compared by log-rank test between the groups. $p < 0.05$ was considered statistically significant.

Results

Patients baseline characteristics

A total of 436 HCC patients from Nanfang Hospital of Southern Medical University between June 2015 to July 2020 that met the inclusion criteria were enrolled for final analyses. Approximately 36.5% (159/436) patients had recurrent tumor, the rest were recurrence-free patients, and they were randomly divided into training and validation cohorts at the ratio of 2:1. The clinical characteristics between patients from training and validation cohorts were detailed in **Table1**, with no statistical difference ($p > 0.05$). The median follow-up period was 46 months (interquartile range, 25 - 81 months) for the training cohort and 55 months (interquartile range, 27 - 84 months) for the validation cohort.

CECT feature evaluation and radiomics signature building

1936 radiomics features were extracted for each HCC patients, and among these first-order features, texture features and wavelet-transformed features, 20 early recurrence-related features including 12 from arterial phase and 8 from portal venous phase were selected by the least absolute shrinkage and selection operator-Cox (LASSO-Cox) method to derive a radiomics signature. The

formula for radiomics signature was shown in **Table S3**. The radiomics signature revealed favorable prediction ability for early recurrence with C index of 0.686 (95% confidence interval [CI]: 0.631 - 0.747) in the training cohort and 0.721 (95% CI: 0.667 - 0.778) in the validation cohort.

Predictive model development and validation

Univariate Cox regression analysis selected tumor size, tumor number, microvascular invasion (MVI) status, alpha-fetoprotein (AFP) level, gamma-glutamyl transpeptidase to lymphocyte ratio (GLR) level and radiomics signature as important factors to evaluate post-operative early recurrence (**Table S4**). Multivariate Cox regression analysis further selected MVI status (hazard ratio [HR]: 1.459, 95% CI: 1.077 - 1.977), serum AFP level (HR: 1.399, 95% CI: 1.022 - 1.915), GLR level (HR: 1.587, 95% CI: 1.148 - 2.195) and radiomics signature (HR: 2.336, 95% CI: 1.643 - 3.323) as independent predictors of HCC early recurrence after surgery, and the four factors were incorporated in the novel combined model to predict 2-year RFS for HCC, with radiomics signature accounting for the highest proportion (**Figure 2**) (formula of radiomics-clinic model was shown in **Table 2**). In training cohort, the radiomics-clinic combined model revealed favorable prediction ability with AUCs of 0.744 (95% CI: 0.687 - 0.801) to predict 2-year RFS (**Figure 3A**); and validation cohort further verified the results with AUCs of 0.821 (95% CI: 0.754 - 0.867) (**Figure 3B**). The calibration curve

suggested that the radiomics-clinic model-predicted RFS was well calibrated with the Kaplan-Meier observed RFS at 2-year time-point (**Figure 3C-D**). Decision curve analysis was applied to determine potential risk threshold for clinical decision and graphically demonstrated the radiomics-clinic model may provide favorable net benefit across the range of reasonable risk threshold probabilities (**Figure 3E-F**).

Radiomics-clinic model for early recurrence risk stratification

Radiomics-clinic scores were calculated for each HCC patient according to the combined model and recorded as their risk scores. Cut-off points were decided on the 33.3% and 66.6% scores of all score range. Finally, HCC patients were differently grouped into three risk groups according to the risk score: patients of score ≤ 1.97 into the low-risk group, 1.97 - 2.30 into the intermediate-risk group, and > 2.30 into the high-risk group (**Table 3**). 2-year RFS difference between different risk-groups was statistically significant tested by log-rank test ($p < 0.001$) (**Figure 4A-B**). And similar results were either observed in sub-group of AFP level $\leq 20\text{ng/ml}$ (**Figure 4C-D**), suggesting our radiomics-clinic model's underlying significance in predicting prognosis for HCC patients with negative AFP level.

Correlation between radiomics features and early recurrence as well as genomic characteristics

Analysis revealed that radiomics signature had statistical significance between the early recurrent patients and

early recurrence-free patients ($p < 0.05$) (**Figure 5A-B**). 48 HCC patients were randomly selected in this analysis, and relative expression of gene transcriptional level of the tumor tissue was assessed by q-PCR method (**Table S5**). Pearson correlation analyses revealed that BCAT1 (PCC = 0.556, $p < 0.001$) and CENPA (PCC = 0.457, $p = 0.002$) had significant association with the radiomics signature (**Figure 5C-D**).

Discussion

HCC is a well-acknowledged malignant tumor known for its heavy tumor burden, high mortality and unfavorable prognosis. As one of the first-line treatment of HCC, hepatic resection is recommended for liver cancer of early stage, however post-operative recurrence rate remains high, and there aren't more effective predictive biomarkers applied in clinic, except for serum AFP level. HCC is stimulated and regulated by polygenes and multi-factors, and is generally developed from chronic liver injuries, inflammation, fibrosis and cirrhosis to tumor^{19,20}. Tumor cell's aggressive biological behavior may be part of the reasons for tumor progression and metastasis, which is one of the important reasons for high post-operative relapse rate and poor prognosis at present. To investigate and learn from malignant tumor behavior, some researchers have probed into the underlying biological code of medical images, and found some links between image features and clinical

outcomes of tumors. In this study, we analyzed the correlations between image features of CECT and HCC patients' early recurrence and prognosis after operation, and figured out four important indicators to construct the novel radiomics-clinic nomogram, with favorable predictive capability.

Tumor heterogeneity exists between and inside the tumor entity, therefore investigators need to learn the tumor regions from medical images more than the information of tumor size, tumor number, etc. ML is one of the important techniques of artificial intelligence; and some researchers applied different approaches to build radiomics signatures^{13,21} in order to learn the texture characteristics of voxel intensity variance and portray the tumor phenotype.

In this research, we decided to learn from CECT images based on the reason that CT was the essential and cost-effective examination before surgery, and contrast-enhanced protocol was important due to the special enhancement patterns of HCC²². Although there were some researchers investigated into the relationship between CT radiomics and patients' prognosis after operation^{12,23,24}, in this study, we further investigated the novel radiomics-clinic model's predictive potential in the AFP-negative HCC cohort and obtained the favorable prediction results. We extracted radiomics features from CECT images and conducted dimensionality reduction applying the LASSO method. The radiomics signature revealed

favorable prediction ability for early recurrence and performed better when incorporating with other clinical predictors. Choice of surveillance strategy should balance sensitivity to optimize early detection of recurrence, specificity to minimize harms from follow-up tests, and costs to remain cost-effective. Patients at high risk of recurrence are potential candidates of adjuvant therapy^{25,26}; individualized risk scoring may provide valuable information for medical reference, and adopting tailor-made post-operative interventions and tighter follow-up visits may benefit HCC patients from longer RFS.

In this study, multivariate Cox regression analysis selected radiomics signature, MVI status, GLR and serum AFP level as significant predictors, and the variables incorporated in the radiomics signature includes 12 features and 8 features from AP and PVP, respectively, with 80.0% (16/20) wavelet transformed features. Based on signal processing and analysis techniques, wavelet transformation is used to analyze the medical images from multiple aspects, and wavelet features frequently account for a large proportion of selected signatures because they capture multi-scale texture heterogeneity, which is often related to tumor aggressiveness, necrosis, or microvascular invasion, although these associations are still exploratory and not mechanistically established, and biological interpretation of such high-order features remains challenging. In a longitudinal assessment study, three serum biomarkers revealed different predictive

ability to detect the very early-stage HCC, and their results suggested that serum AFP level alone had the most favorable discriminating power and the power increased when combined with AFP-L3; and the AFP and AFP-L3 combination greatly improved the sensitivity for detecting HCC at the very early stage compared to ultrasonography test²⁷. Though tumor size and number, capsular invasion, portal vein invasion, as well as MVI were significant predictors for HCC recurrence, more importantly, serum AFP level was considered as the most representative biomarker for post-operative recurrence, and the results suggested baseline AFP and albumin level as important relapse-related predictors²⁸. One study reported that occurrence rate of pathologically proved MVI was 15.0% to 57.1% in surgically resected liver cancer specimen²⁹. MVI being the malignant manifestation of HCC, is one independent risk factor for tumor recurrence and always associated with poor prognosis. Some investigations also studied the relationship of radiomics features and the presence of MVI, as well as constructed radiomics models to predict MVI status^{7,30,31}, however in our study, MVI status acts as an independent predictor of early recurrence. Furthermore, infiltration of inflammatory factors may promote angiogenesis and tumor progression³². In our and other previous studies, GLR was identified as a novel underlying factor for HCC's

prognosis prediction^{33,34}. Therefore, the predictive potential of our radiomics-clinic model increased when incorporating these significant predictors with the radiomics signature.

Radio-genomics is another emerging field which aims to study the association between radiomics features and genomic characteristics, and therefore provides genetic evidence for radiomics analyses and may figure out important therapeutic targets. Segal. et al were the first to investigate the correlation between HCC's gene expression pattern and the corresponding imaging features in 2007, however, the features in their study were radiological features³⁵. Followed by this, many investigators either probed into the relationships³⁶. One study interpreted the association of radiomic phenotypes with biological pathways of p53 mutant HCC, and therefore offered biologic basis of the underlying mechanism of individual radiomics phenotypes' association with survival outcome for HCC³⁷. Moreover, in some recent studies, BCAT1^{38,39} and CENPA⁴⁰ were identified as important biomarkers for HCC post-operative surveillance and early recurrence as well as prognosis prediction. In our study, we also found correlation between these two genes and CT radiomics signature, which to some extent, may offer reference to comprehend medical image's role in understanding tumor biology.

There are some limitations in this study. Firstly, this is a retrospective study with inherent bias in patient

enrollment, therefore in future study, perspective cohort is needed for further verification. Secondly, our data were received from a single medical center, and validations in multiple centers from different countries or regions may offer more adequate evidence for clinical application of the novel combined model. Thirdly, the biological significance of radiomics signature as well as a single feature is still poorly understood, posing a barrier in application of radiomics, and this may be an important direction for more future studies. Fourthly, the radio-genomic analysis was designed as an initial exploratory step in the current study, and further mechanistic and functional experiments might provide preliminary biological context for the radiomics findings.

Conclusion

In conclusion, our study constructed a novel prediction model based on CT radiomics signature and clinical features, and the model performed well in AFP-negative HCC cohort, which may provide potential support for pre-operative prediction of early recurrence risk and offer guidance for post-operative surveillance. And this study may promote the practical application of artificial intelligence technology in clinical practice.

Figure legends

Figure 1. HCC patients' enrollment diagram.

Figure 2. Construction of the CT radiomics-clinic

combined model.

The total score was calculated on the basis of radiomics signature (rad-score), microvascular invasion (MVI), gamma-glutamyl transpeptidase to lymphocyte ratio (GLR) and alpha-fetoprotein (AFP), and model predicted early recurrence rate was determined corresponding to the total score.

Figure 3. Radiomics-clinic combined model predicted early recurrence.

Receiver operating characteristic (ROC) curve analysis was performed in the training (A) and validation (B) cohorts to assess the nomogram's ability in predicting early recurrence. Calibration curves of 2-years RFS in training cohort (C) and validation cohort (D). Decision curve analysis of the nomogram in training cohort (E) and validation cohort (F). AUC, area under the ROC curve; RFS, recurrence-free survival.

Figure 4. RFS of three risk groups defined by the radiomics-clinic model.

Kaplan-Meier plots for RFS in the low-, intermediate- and high-risk groups as defined by the radiomics-clinic model in training cohort (A) and validation cohort (B). And similar results were shown in sub-group of AFP level ≤ 20 ng/ml (C-D). Median RFS, hazard ratios (with p values) and percentage RFS at 2 years, were reported in Table 3. RFS, recurrence-free survival; AFP, alpha-fetoprotein.

Figure 5. Correlation of radiomic features with early recurrence and gene characteristics.

Radiomic features comparison between patients with or

without early recurrence in the training cohort (A) and validation cohort (B). The mRNA levels of genes BCAT1 (C) and CENPA (D) were associated with radiomic features.

Abbreviations

HCC Hepatocellular carcinoma

BCLC Barcelona Clinic Liver Cancer

TME Tumor microenvironment

AI Artificial intelligence

ML Machine learning

DL Deep learning

RFS Recurrence-free survival

AP Arterial phase

PVP Portal venous phase

ROI Region of interest

LASSO Least absolute shrinkage and selection operator

AIC Akaike Information Criteria

ROC Receiver operating characteristic curve

AUC Area under the curve

DCA Decision curve analysis

Q-PCR Quantitative real-time PCR

PCC Pearson Correlation Coefficient

SD Standard deviation

IQR Interquartile range

MVI Microvascular invasion

AFP Alpha-fetoprotein

GLR Gamma-glutamyl transpeptidase to lymphocyte ratio

HR Hazard ratio

CI Confidence interval

Data availability

The datasets used and analyzed during the current study are available from the corresponding author upon reasonable request.

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Authors' contributions

H.R., Y.Z. and F.L. put forward and designed the study; M.L., N.L. and S.H. wrote this article and participated in the collection and analysis of data. X.W., Z.W., B.L., X.A. and K.W. participated in the analysis of data and help revised the manuscript. All authors contributed to the article and approved the final manuscript.

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Declarations**Competing interests**

The authors declare that they have no competing interests.

Ethics approval and consent to participate

This study was approved by the Ethics Committees of the Nanfang Hospital of Southern Medical University (NFEC-202305-K32), and written informed consent was waived by the Ethics Committee for the retrospective nature of this study

Consent for publication

All authors read and approved the final manuscript.

Patients underwent hepatic resection with liver cancer were collected from the center from 6/2015 to 7/2020 (n = 664)

Exclusion criteria

Patients were excluded (n = 228)

1) non-radical surgery (n = 27)

2) postoperative pathological diagnosis as non-HCC (n = 99)

3) not the first primary cancer (n = 39)

4) died in two month after operation (n = 3)

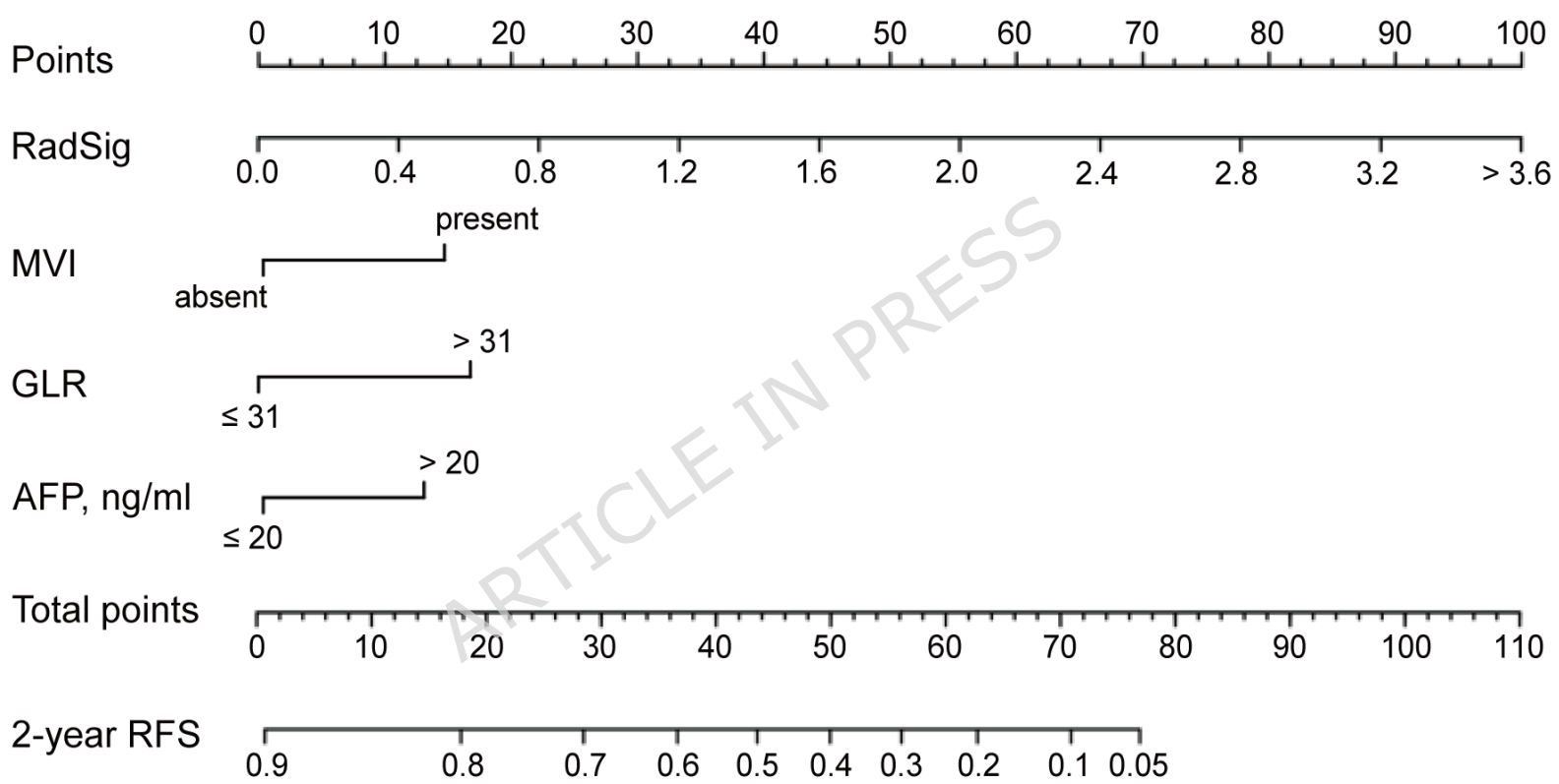
5) images with significant artifact (n = 31)

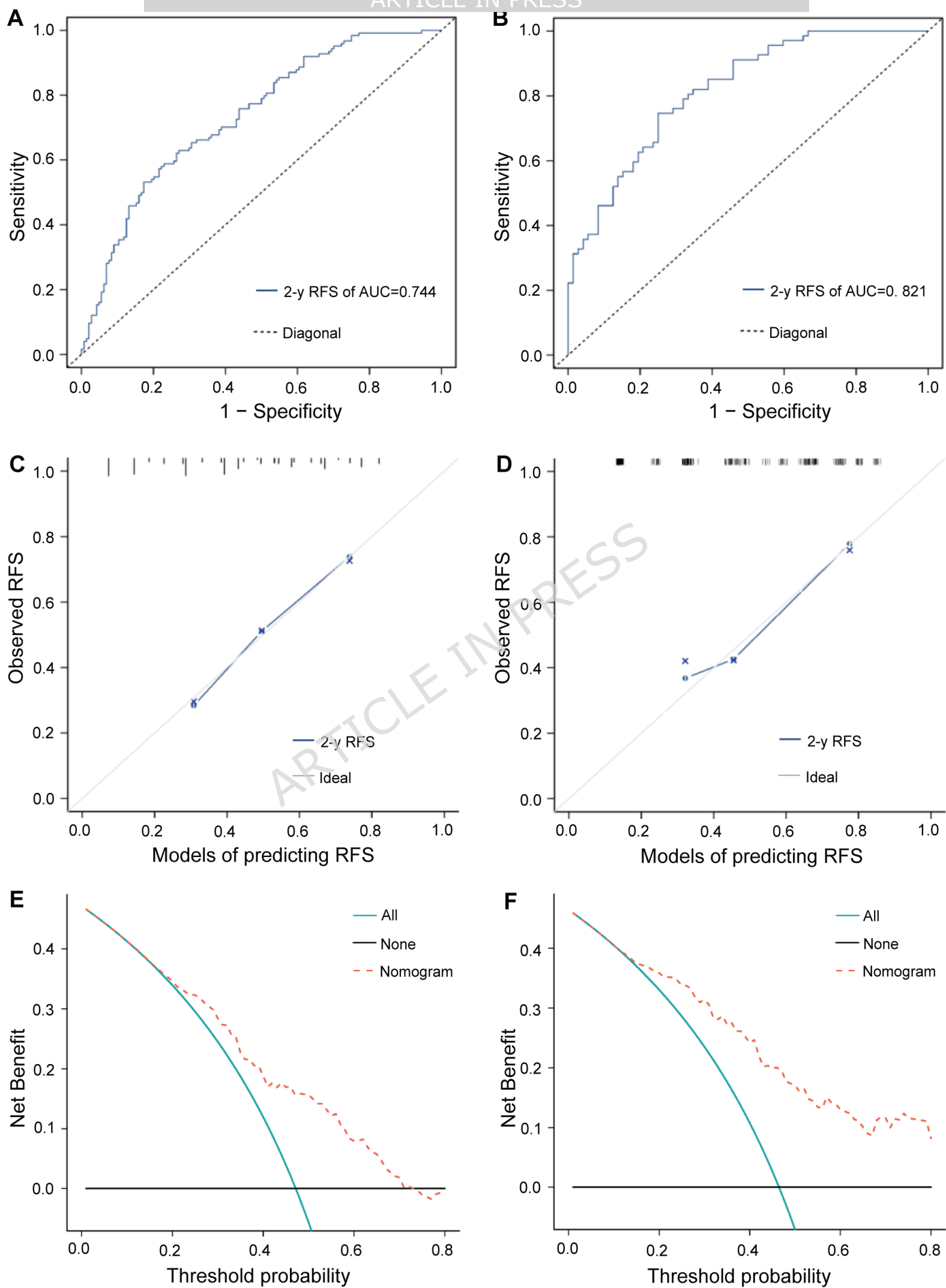
6) data is incomplete, or lost contact in follow-ups (n = 29)

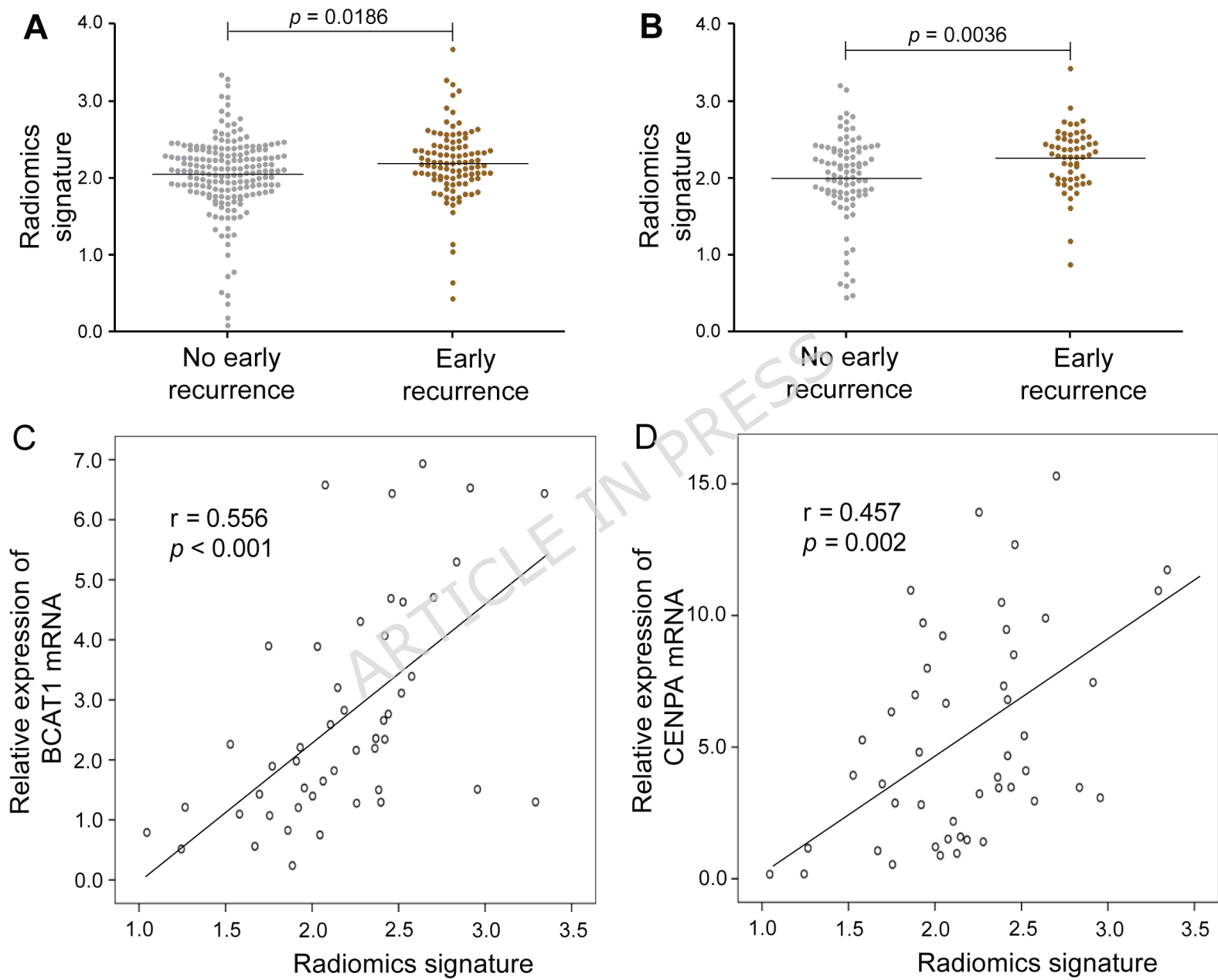
Totally 436 patients were included

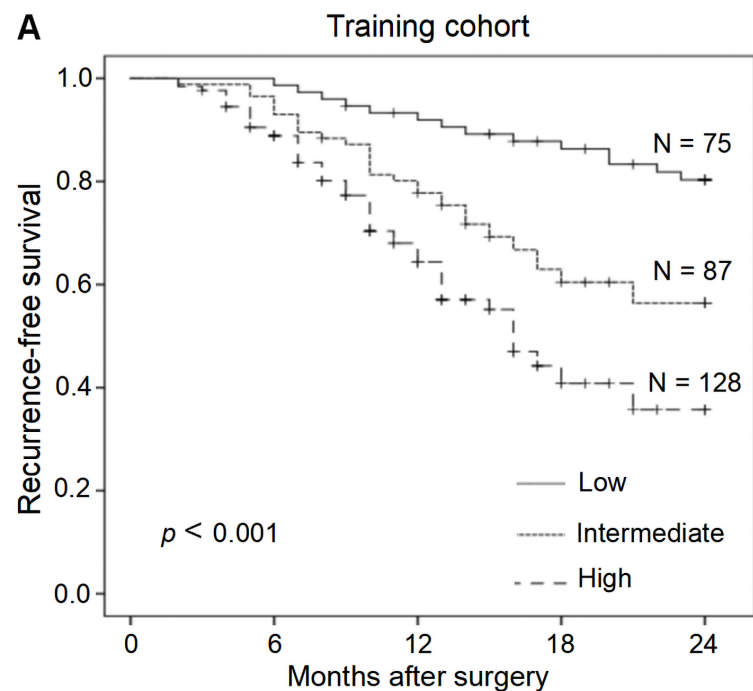
Training cohort (n = 290)

Validation cohort (n = 146)



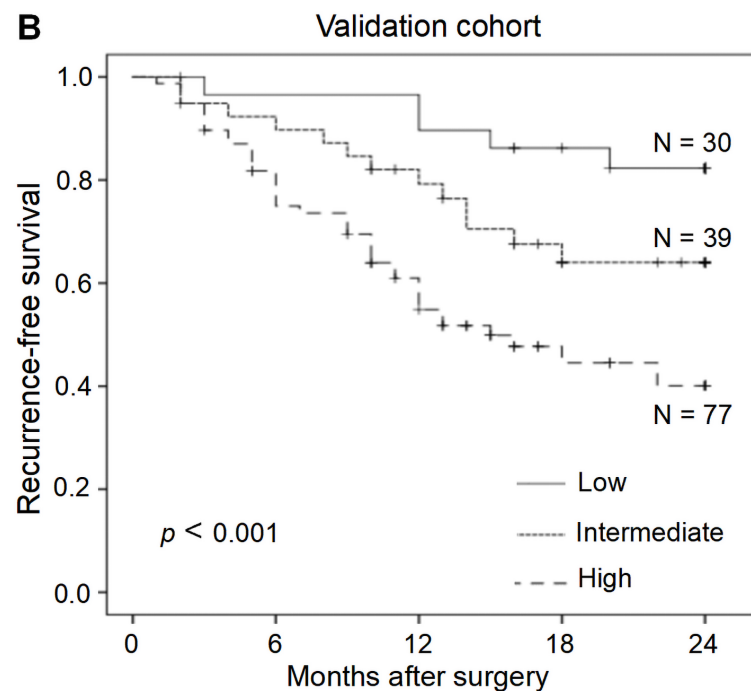






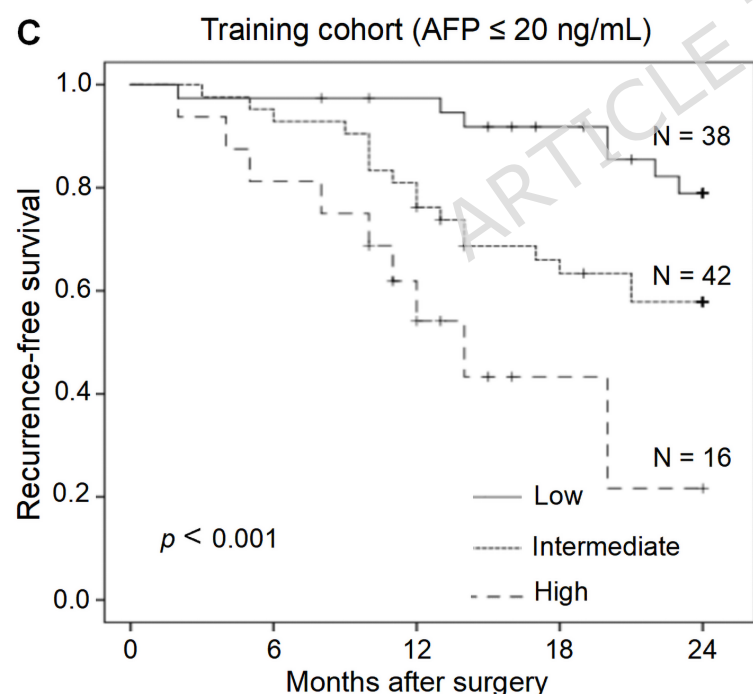
Number at risk: N (%)

Low	75(100)	74(99)	67(89)	59(79)	51(68)
Intermediate	87(100)	82(94)	67(77)	49(56)	40(46)
High	128(100)	110(86)	56(44)	13(10)	5(4)



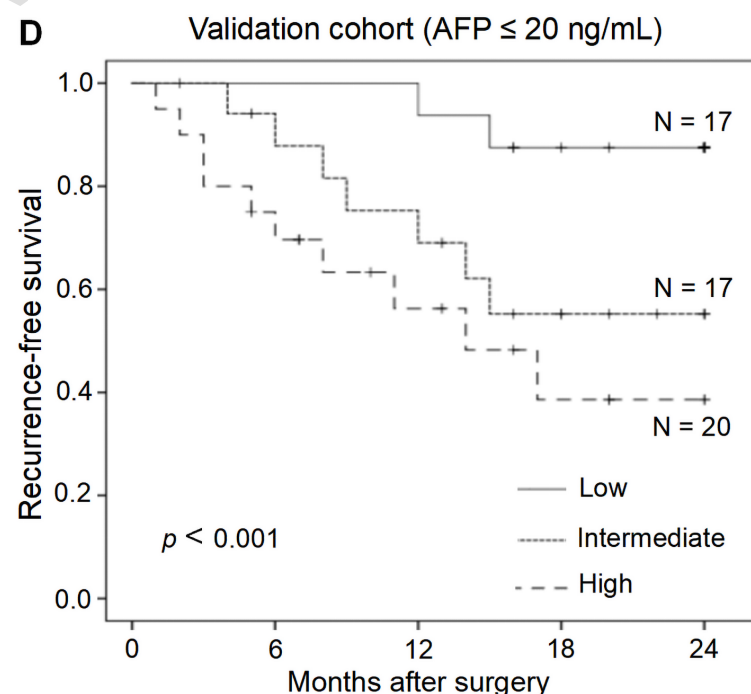
Number at risk: N (%)

Low	30(100)	28(93)	27(90)	22(73)	19(63)
Intermediate	39(100)	35(90)	28(72)	18(46)	11(28)
High	77(100)	58(75)	38(49)	13(17)	7(9)



Number at risk: N (%)

Low	38(100)	37(97)	35(92)	30(79)	23(61)
Intermediate	42(100)	39(93)	33(79)	24(57)	20(48)
High	16(100)	12(75)	7(44)	2(13)	1(6)



Number at risk: N (%)

Low	17(100)	16(94)	15(88)	11(65)	9(53)
Intermediate	17(100)	14(82)	11(65)	6(35)	2(12)
High	20(100)	13(65)	7(35)	4(20)	2(10)

Table 1. Comparison of clinicopathological characteristics of two cohorts' patients.

Parameter	Total patients (n = 436)	Training cohort (n = 290)	Validation cohort (n = 146)	<i>p</i> value
Gender: female/male (n)	63/373	40/250	23/123	0.58 3
Age (years)	51.99±11.20	52.46±11.11	51.06±11.35	0.22 0
HBsAg: negative/positive (n)	73/363	45/245	28/118	0.33 4
Family history: no/yes (n)	396/40	267/23	129/17	0.20 5
Alcohol abuse: no/yes (n)	255/181	161/129	94/52	0.07 6
Smoking: no/yes (n)	260/176	172/118	88/58	0.84 7
Cirrhosis: no/yes (n)	34/402	22/268	12/134	0.81 6
Tumor size (cm)	7.36±4.19	7.16±4.04	7.75±4.47	0.15 2
TNb: single/multiple (n)	345/91	227/63	118/28	0.53 7
MVI: no/yes (n)	253/183	171/119	82/64	0.57 6
Child stage: A/B (n)	400/36	263/27	137/9	0.26 0
BCLC stage: 0-A/B-C (n)	245/191	161/129	84/62	0.68 9
Early recurrence: no/yes (n)	277/159	187/103	90/56	0.56 1
WBC (×10 ⁹ /L)	6.49±2.08	6.53±2.14	6.42±1.96	0.60 2
NEUT (×10 ⁹ /L)	3.93±1.87	3.93±1.93	3.94±1.75	0.98 7
LYMPH (×10 ⁹ /L)	1.71±0.61	1.70±0.63	1.72±0.57	0.73 4
Platelets (×10 ⁹ /L)	194.13±85.96	191.69±80.56	198.91±95.80	0.40 8
Albumin (g/L)	38.37±4.49	38.50±5.12	38.15±4.60	0.47 2
Globulin (g/L)	31.68±6.05	31.59±5.86	31.84±6.41	0.68 5
TBIL (μmol/L, median,	12.64	12.88	12.18	0.05

IQR)	(9.21-16.87)	(9.38-17.46)	(8.96-16.01)	5
DBIL (μmol/L, median, IQR)	5.0 (3.81-5.9)	5.0 (3.81-5.9)	5.0 (3.8-5.9)	0.27 2
ALT (U/L, median, IQR)	32.11 (20.9-50.47)	32.44 (21.41-46.85)	31.6 (19.43-55.2)	0.14 4
AST (U/L, median, IQR)	35.83 (26.42-54.12)	36.0 (26.27-53.3)	35.4 (26.8-56.5)	0.50 3
GGT (U/L, median, IQR)	75.49 (44.21-130.23)	45.25 (73.0-125.47)	78.05 (42.73-145.8)	0.40 7
AFP (ng/ml, median, IQR)	157.5 (8.08-1210.0)	162.9 (8.82-1210.0)	145.9 (6.66-1210.0)	0.58 5
GLR level (median, IQR)	45.81 (24.51-84.31)	45.82 (24.52-80.39)	45.12 (23.65-97.95)	0.81 5
Radiomics signature	2.13±0.47	2.15±0.46	2.11±0.50	0.43 9

Abbreviations: n, number of patients; HBsAg, hepatitis B surface antigen; TNb, tumor number; MVI, Microvascular invasion; BCLC, tumor-node-metastasis; WBC, white blood cell; NEUT, neutrophil count; LYMPH, lymphocyte count; TBIL, total bilirubin; IQR, Interquartile range; DBIL, direct bilirubin; ALT, alanine aminotransferase; AST, aspartate aminotransferase; GGT, Gamma-glutamyl transpeptidase; ALP, alkaline phosphatase; AFP, alpha-fetoprotein; GLR, GGT to LYMPH ratio.

Table 2. Multivariable Cox regression analyses of early-recurrence factors in the training cohort.

Variable	β -estimate (95% CI)			HR (95% CI)			<i>p</i> value
Tumor size, cm (> 5 <i>vs.</i> ≤ 5)	0.002	(-0.338	to 0.342)	1.002	(0.713	to 1.408)	0.991
TNb (multiple <i>vs.</i> single)	0.037	(-0.318	to 0.391)	1.038	(0.728	to 1.478)	0.838
MVI (present <i>vs.</i> absent)	0.378	(0.074	to 0.682)	1.459	(1.077	to 1.977)	0.015
AFP, ng/ml (> 20 <i>vs.</i> ≤ 20)	0.336	(0.022	to 0.650)	1.399	(1.022	to 1.915)	0.036
GLR level (> 31 <i>vs.</i> ≤ 31)	0.462	(0.138	to 0.787)	1.587	(1.148	to 2.195)	0.005
Radiomics signature	0.849	(0.497	to 1.202)	2.336	(1.643	to 3.323)	< 0.001

Model score = $0.378 \times \text{MVI}$ (0: absent, 1: present) + $0.336 \times \text{AFP}$ (ng/ml) (0: ≤ 20; 1: > 20) + $0.462 \times \text{GLR}$ (0: ≤ 31; 1: > 31) + $0.849 \times \text{Radiomics signature}$.

Cut-offs to generate the risk groups: ≤ 1.97 (low), > 1.97 to ≤ 2.30 (intermediate), > 2.30 (high)

Abbreviations: HR, hazard ratio; CI, confidence interval; TNb, tumor number; MVI, microvascular invasion; AFP, alpha-fetoprotein; GLR, gamma-glutamyl transpeptidase to lymphocyte ratio.

Table 3. Median RFS, hazard ratio and 2-year RFS of different risk group as defined by model.

Cohort	Group	n	Median RFS, months (95% CI)	Hazard ratio (95% CI)	<i>p</i> value	2-year RFS (%)
Training cohort	Low	75	61.73 (54.71-68.74)	1		80.26
	Intermediate	87	42.33 (35.48-49.49)	3.43 (1.65-7.16)	0.001	55.17
	High	12 8	25.96 (21.85-30.06)	6.33 (3.17-12.65)	< 0.001	36.11
Validation cohort	Low	30	69.60 (60.03-79.11)	1		83.03
	Intermediate	39	50.10 (39.81-60.51)	3.12 (0.983-9.93)	0.048	64.15
	High	77	28.40 (22.27-34.58)	4.28 (1.48-12.35)	0.005	39.97
All	Low	10 5	64.48 (58.61-70.29)	1		81.17
	Intermediate	12 6	44.70 (38.96-50.47)	3.33 (1.79-6.19)	< 0.001	57.86
	High	20 5	26.90 (23.45-30.36)	5.41 (3.05-9.66)	< 0.001	37.64

RFS, recurrence-free survival; CI, confidence interval.